

RESEARCH PROJECT

NASA Near-Earth Asteroids Dataset

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CS3120

Initial Cleanup

Data Preparation



```
1 # Columns to drop: identifiers, raw URLs, redundant text, and near-empty columns
2 COLS_TO_DROP = [
3     'id', 'neo_id', 'name', 'short_name', 'designation',
4     'nasa_url', 'api_url',          # identifiers, not directly useful
5     'close_approach_data',         # raw JSON string - would need separate parsing
6     'sentry_data',                 # almost entirely empty, link to API
7     'orbit_class_desc', 'orbit_class_range', # duplicated by orbit_class_type
8     'equinox',                     # constant value (J2000) across all rows
9     'epoch',                        # raw date, not directly useful
10    'perihelion_time',               # highly correlated with other orbital params
11    'sentry',                        # nearly all false except for objects marked hazardous
12 ]
13
14 df_clean = df.drop(columns=COLS_TO_DROP)
```

```
1 # Extract the year of first observation as a feature if the column exists
2 if 'first_observation_date' in df_clean.columns:
3     df_clean['first_obs_year'] = pd.to_datetime(df_clean['first_observation_date']).dt.year
4     print("New feature: 'first_obs_year' created.")
5
6 # Drop the raw date columns safely using errors='ignore'
7 date_cols_to_drop = [
8     'first_observation_date',
9     'last_observation_date',
10    'orbit_determination_date'
11 ]
12 df_clean.drop(columns=date_cols_to_drop, errors='ignore', inplace=True)
13
14 print("Date column processing complete.")
15
```

```
4 # One-hot encode, drop_first avoids the dummy variable trap
5 df_clean = pd.get_dummies(df_clean, columns=['orbit_class_type'], drop_first=True)
6
7 print(f"\nShape after encoding: {df_clean.shape}")
8 df_clean.head(3)
9
```

Exploratory Data Analysis

Feature Overview: 2000 Objects, 24 Features

TARGET FEATURE

potentially_hazardous — Boolean flag indicating whether the asteroid poses a theoretical collision risk with Earth

perihelion_distance

Closest approach to the Sun (AU)

aphelion_distance

Farthest point from the Sun (AU)

inclination

Orbital tilt relative to the plane of the ecliptic

min_orbit_intersection

Closest Earth approach at orbit crossing (AU)

diameter_min_m / max_m

Estimated size bounds in meters

magnitude

Brightness (inversely related to size)

orbit_uncertainty

Confidence score for the computed orbit

orbital_period

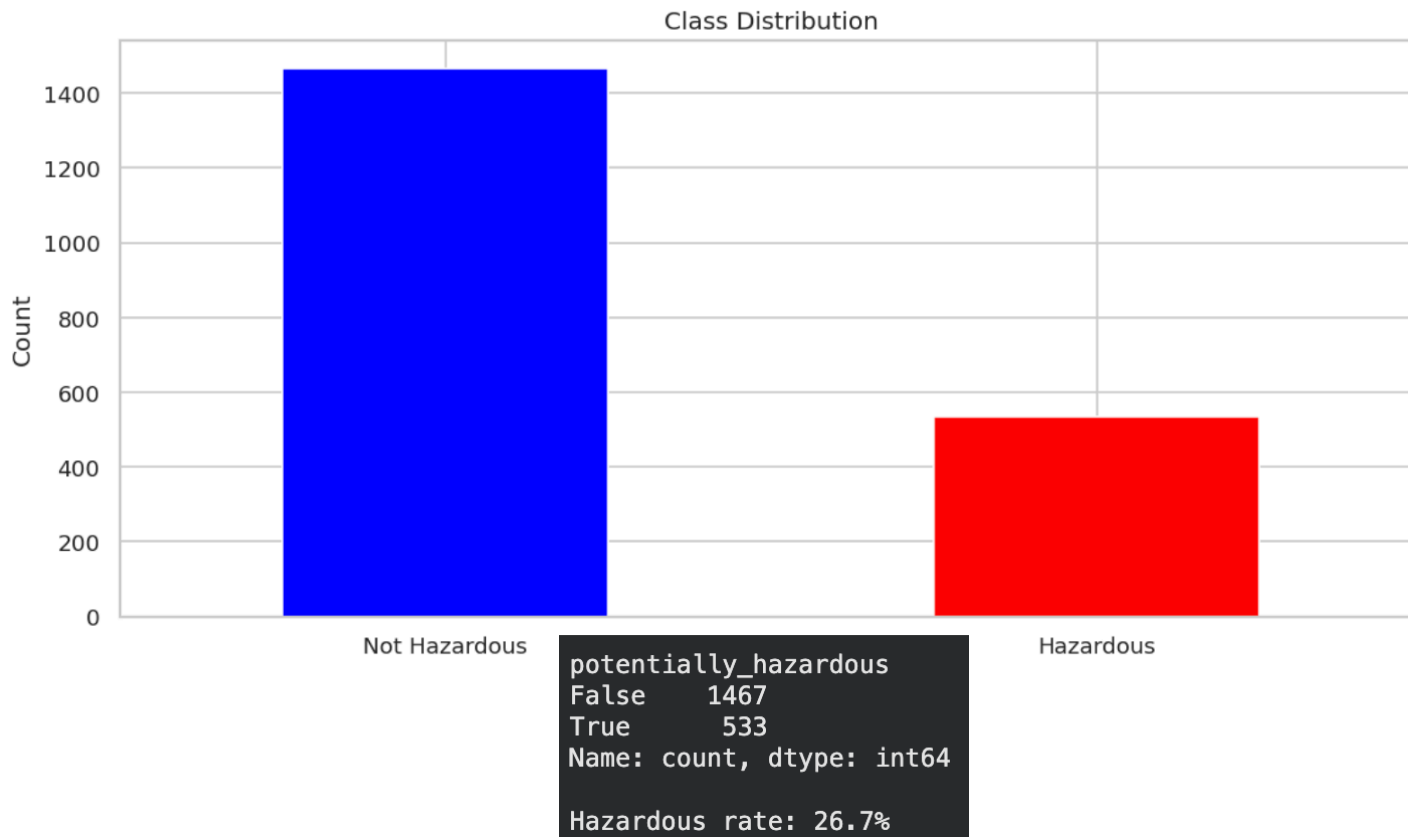
Days to complete one orbit around the Sun

orbit_class_type

Apollo, Amor, Aten, or IEO classification

Exploratory Data Analysis

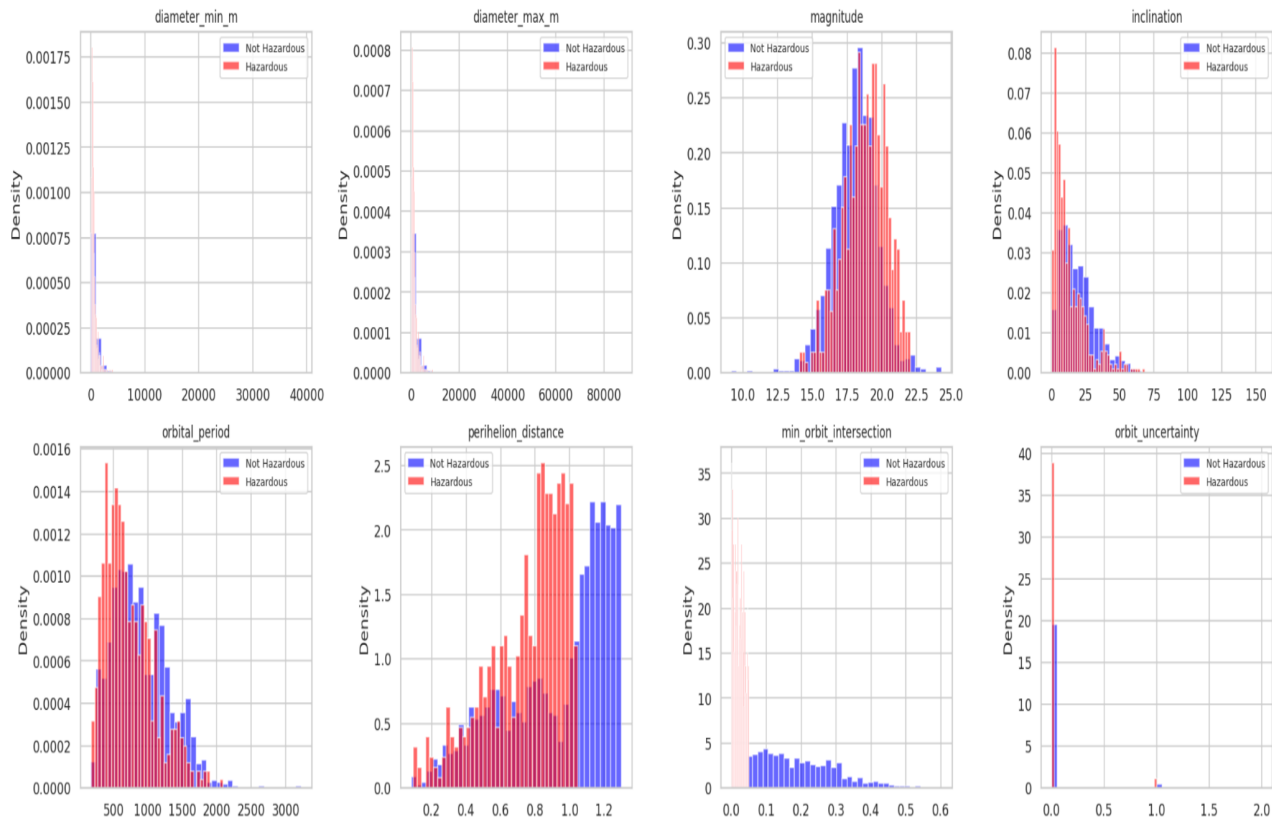
Class distribution based on target feature



Exploratory Data Analysis

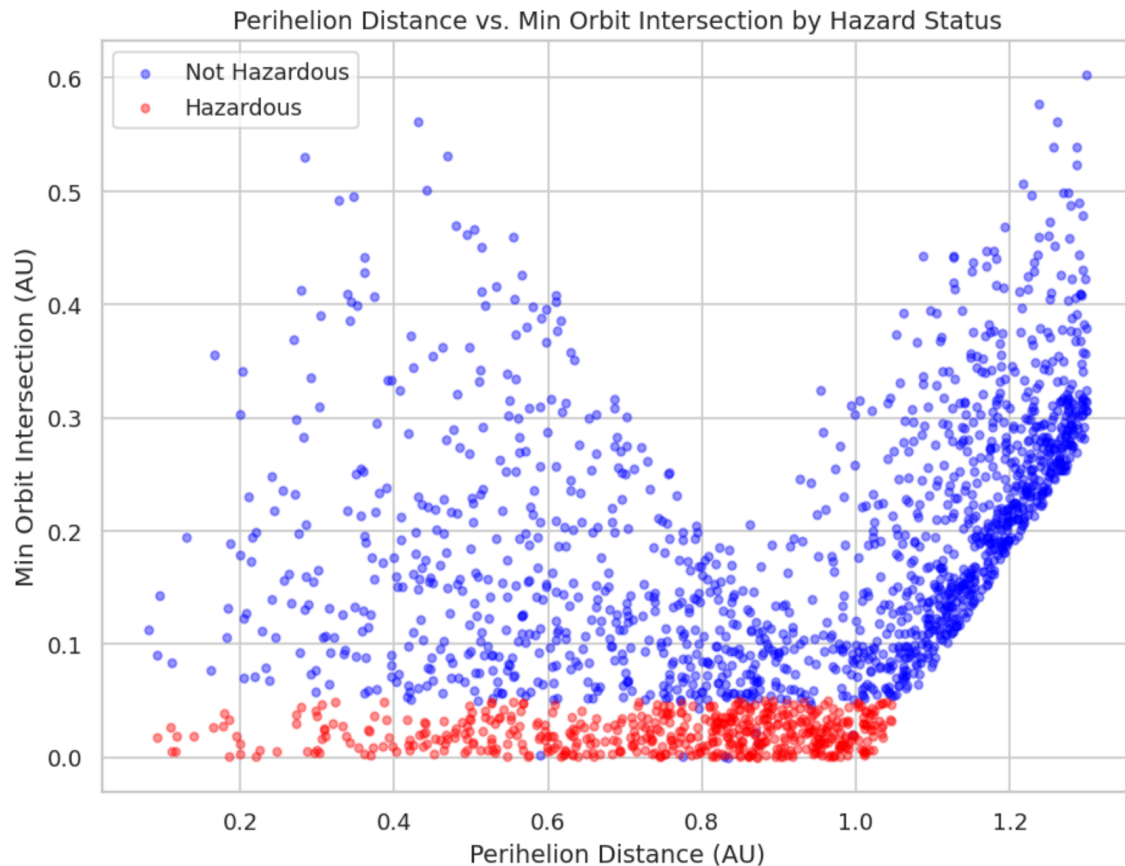
Feature Distributions · Hazardous vs. Non-Hazardous Comparison

Feature Distributions: Hazardous vs. Not Hazardous



Exploratory Data Analysis

Perihelion Distance vs. Min-Orbit Intersection by Hazard Status

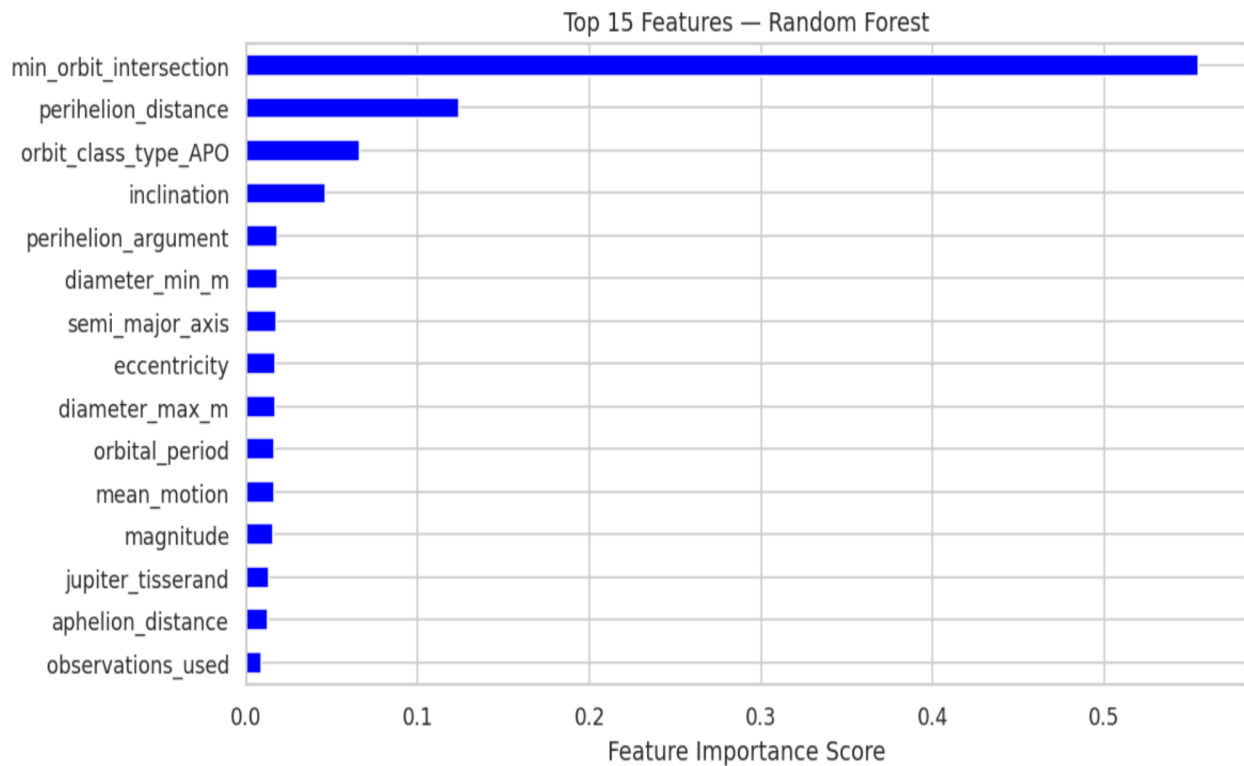


Hypothesis Testing

H1 · Orbital Mechanics & Hazard Designation

H1

Smaller perihelion distance, min_orbit_intersection, and inclination will each be significantly associated with the potentially hazardous designation.

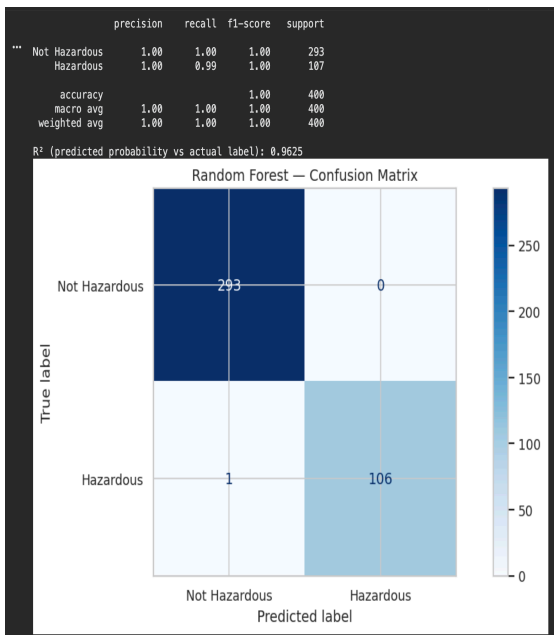


Hypothesis Testing

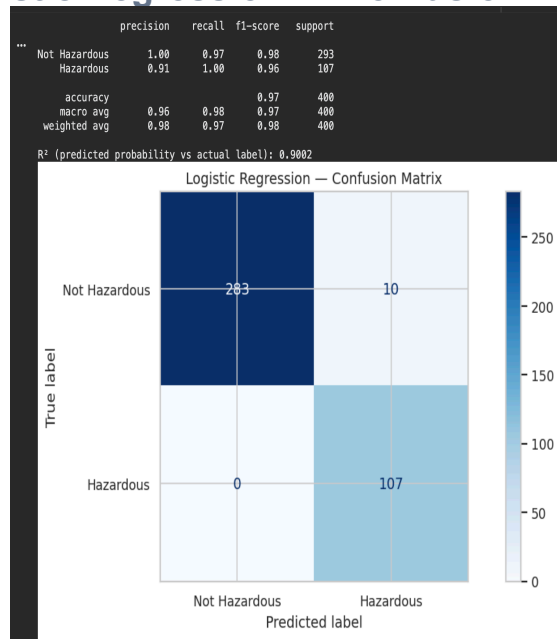
H2 · Classification Model Performance

H2 A classification model trained on the given features will achieve perfect recall for PHOs, because the PHO designation derives from these same measurable orbital quantities.

Random Forest — Confusion Matrix



Logistic Regression — Confusion Matrix



Conclusion

Key Takeaways

Asteroid hazard classification is not a black box.

It is driven by a handful of well-understood orbital mechanics, and these models validated the classification with very strong performance.

A key tension between precision and recall was revealed.

In planetary defense, missing a hazardous object is far more costly than a false alarm. This tradeoff should guide how future classification models are tuned and deployed.

AI Tool Usage

Tools used in the development of this project

Claude Sonnet 4.6 Adaptive

Usage

- Guided exploratory data analysis
- Explained feature definitions in context
- Assisted with model training strategy

Gemini 3.1 Pro

Usage

- Bug fixing inside Google Colab